

BROKEN BOW, NE, USA

WMO: 725555

Lat: 41.433N

Lon: 99.633W

Elev: 2530

StdP: 13.40

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 94946

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.2	1.3	-10.2	3.5	-2.1	-5.9	4.4	2.9	35.2	31.6	32.2	30.2	7.5	340	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.8	94.5	70.6	90.9	71.0	88.2	70.5	75.9	87.0	74.3	85.0	72.7	83.4	13.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.5	132.5	81.9	71.0	125.8	80.8	69.6	119.4	79.5	41.2	87.2	39.6	84.7	38.0	83.0	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
30.7	26.4	23.5	DB	-12.2	98.9	4.2	3.6	-15.2	101.4	-17.6	103.5	-20.0	105.5	-23.0	108.1	
			WB	-12.6	79.0	4.1	2.1	-15.6	80.5	-17.9	81.8	-20.2	83.0	-23.2	84.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.5	26.4	27.0	38.8	48.4	58.6	69.2	74.6	71.7	63.5	49.8	37.5	26.9
	DBStd	19.72	11.71	12.14	11.04	9.71	8.69	6.86	5.44	5.59	8.42	9.30	10.60	11.14
	HDD50	3136	732	644	372	140	23	0	0	0	5	116	389	715
	HDD65	6471	1196	1063	813	502	235	36	5	11	128	477	825	1180
	CDD50	2950	0	1	24	91	289	576	762	673	410	110	14	0
	CDD65	813	0	0	1	3	37	162	302	219	83	6	0	0
	CDH74	9186	0	0	30	121	535	1790	3282	2232	1031	154	11	0
	CDH80	3735	0	0	5	25	169	684	1509	918	380	44	1	0
Wind	WSAvg	9.8	10.2	10.4	11.0	12.3	11.1	9.8	7.8	7.6	8.7	9.5	9.6	9.9
Precipitation	PrecAvg	22.6	0.4	0.5	1.3	2.3	3.3	3.9	3.4	2.5	2.1	1.3	0.9	0.4
	PrecMax	35.2	1.6	1.7	5.8	6.0	7.3	10.3	9.1	7.8	6.0	3.6	2.8	1.1
	PrecMin	11.3	0.0	0.0	0.0	0.2	0.7	1.0	0.2	0.1	0.2	0.0	0.0	0.0
	PrecStd	4.1	0.4	0.4	1.1	1.4	1.7	2.3	1.9	1.5	1.6	1.0	0.7	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.4	65.7	78.4	83.6	90.5	96.6	99.5	97.2	92.9	86.2	74.7	60.8
		MCWB	45.0	48.6	55.1	60.1	64.1	69.5	71.4	70.8	67.0	61.6	53.1	44.9
	2%	DB	53.8	57.1	70.2	77.6	84.6	91.1	95.4	92.3	88.9	78.1	67.8	53.6
		MCWB	41.3	44.0	51.3	57.4	64.5	70.4	71.0	71.0	68.4	58.8	50.3	41.5
	5%	DB	47.9	51.1	63.8	72.1	80.2	87.6	91.4	88.6	84.3	72.7	61.3	47.7
		MCWB	38.3	40.7	49.9	55.2	62.9	70.0	72.5	71.5	66.9	56.0	47.4	38.6
	10%	DB	43.0	45.2	57.7	66.5	75.2	83.5	88.4	85.1	80.4	67.0	54.8	42.9
		MCWB	35.7	37.8	46.2	52.6	60.8	68.1	72.2	71.1	64.4	53.5	44.5	35.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.6	50.8	59.1	62.3	70.1	76.0	78.4	77.1	73.5	64.5	55.6	47.1
		MCDB	60.7	62.3	67.3	76.3	81.6	87.5	89.5	86.8	84.1	76.9	69.6	57.0
	2%	WB	41.7	45.4	54.9	59.3	67.3	73.4	76.5	75.3	70.9	61.3	51.6	41.9
		MCDB	52.8	56.0	65.2	72.6	79.6	84.4	87.9	85.9	82.8	72.0	63.3	52.6
	5%	WB	38.9	41.5	50.5	56.6	64.8	71.7	75.0	73.4	68.9	58.4	48.4	38.8
		MCDB	47.0	51.0	63.2	69.6	76.6	83.9	86.0	83.3	80.3	68.9	60.5	46.8
	10%	WB	36.2	37.9	46.6	53.7	62.4	69.7	73.3	71.8	66.8	55.2	44.6	35.7
		MCDB	42.4	44.8	57.9	64.7	72.8	81.4	84.0	81.8	77.0	65.3	55.0	42.1
Mean Daily Temperature Range	5% DB	MDBR	22.1	22.6	25.2	24.7	23.2	23.3	23.8	24.0	26.8	26.5	25.9	21.9
		MCDBR	30.3	32.2	36.6	35.0	30.4	28.0	29.2	29.1	30.8	36.2	37.0	29.3
		MCWBR	19.6	20.5	21.0	18.4	14.5	12.2	12.0	12.5	13.5	18.6	21.6	19.4
	5% WB	MCDBR	28.3	30.6	33.6	30.6	26.2	24.2	23.9	23.7	26.1	30.0	33.9	28.2
MCWBR		18.9	19.9	20.1	17.0	13.8	12.2	12.2	12.2	12.9	17.3	20.7	19.3	
Clear-Sky Solar Irradiance	taub	0.268	0.273	0.297	0.328	0.356	0.371	0.389	0.384	0.351	0.313	0.273	0.263	
	taud	2.500	2.502	2.466	2.396	2.401	2.389	2.381	2.377	2.426	2.511	2.540	2.508	
	Ebn at Noon	286	302	304	299	292	286	279	278	281	282	283	279	
	Edh at Noon	23	27	31	36	37	37	37	36	32	26	22	21	
All-Sky Solar Radiation	RadAvg	698	987	1314	1642	1908	2198	2187	1886	1526	1080	764	603	
	RadStd	59	60	113	85	136	146	77	91	92	111	55	47	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page